

Using Lunar Gravity Assists to Boost Earth Escape for the ESA Comet Interceptor Mission

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Abstract – The ESA Comet Interceptor mission, planned for launch in 2028-2029, will fly by a dynamically new long-period comet from a quasi-halo parking orbit around the Sun–Earth L_2 point. Because the target is unknown at launch, Earth escape must be executed rapidly and cheaply once a comet is identified. The Earth-bound unstable manifold of the L_2 halo orbit naturally reaches the Earth–Moon system, so a lunar gravity assist can raise the escape velocity and reduce the heliocentric injection cost. This work quantifies that benefit by formulating both the lunar-flyby and the direct-escape strategies within the same three-dimensional Sun–Earth CR3BP framework and comparing them, target by target, through ΔV –time-of-flight Pareto fronts. Seven comets are analysed: four sampling the representative encounter geometries (Earth-leading/trailing, $R_c \leq 1$ AU), and three ESA Dry-Run targets. The results show that the swingby benefit is governed by whether the comet crosses the ecliptic inside the corridor swept by the manifold. Targets crossing outside that corridor benefit markedly: the flyby is either strictly necessary or lowers the minimum required ΔV by nearly 30%; those crossing inside it gain nothing. An energy-based analysis removing the propulsive contribution inside a 0.2 AU control radius validates the escape velocities against previous work.

I. Introduction and State of the Art

Comets are among the most pristine remnants of the early Solar System, yet every comet visited so far has completed multiple perihelion passages, progressively altering its composition and morphology [1]. The ESA Comet Interceptor mission, selected in 2019 as the first F-class fast-track mission under the Cosmic Vision Programme, addresses this gap by targeting a dynamically new or long-period comet (LPC) on its first approach to the inner Solar System [2]. Scheduled for launch in 2028-2029, the spacecraft will be deployed into a quasi-halo orbit around the Sun–Earth L_2 Lagrange point,

where it will remain in a waiting phase of up to three years until a suitable target is identified [3].

A. Theoretical and Computational Background

Before addressing the specific problem of designing escape trajectories from a Sun–Earth L_2 halo orbit, it is useful to review the theoretical and computational tools that underpin modern trajectory design in multi-body environments. Much of the dynamical behaviour exploited here is captured by the Circular Restricted Three-Body Problem (CR3BP), which models a massless spacecraft under the gravity of two primaries in circular orbit about their barycenter and has long provided a simplified yet powerful framework for low-energy design near the libration points [4, 5, 6, 7]. These libration (Lagrange) points, where gravitational and centrifugal forces balance in the rotating frame, host families of periodic orbits — Lyapunov, Halo and Lissajous — whose computation by numerical and analytical continuation methods is by now well established [8, 9, 10]; halo orbits are especially attractive for their favourable observational geometry and accessibility. Every such periodic orbit possesses associated stable and unstable invariant manifolds [4], which act as natural transport channels in phase space: Koon et al. [11] demonstrated the existence of heteroclinic connections between the L_1 and L_2 orbits, and these manifold “tubes” form the basis of the Interplanetary Transport Network [4]. Poincaré maps reduce the continuous dynamics to intersections with a lower-dimensional section, exposing the invariant structures and manifold intersections required to construct feasible transfers [12]. Beyond the natural dynamics, gravity-assist manoeuvres provide propellant-free changes of the velocity vector and are a cornerstone of interplanetary mission design [13]. Trajectory design is therefore posed as an optimization problem: gradient-based methods from optimal control are efficient when good initial guesses are available, but the strongly nonlinear multi-body landscape, riddled with local minima, has motivated global techniques such as genetic algorithms [14, 15] and hybrid schemes coupling

*All code to reproduce the results presented in this work, and to produce new ones, together with explanations and tutorials, is publicly available on GitHub at: https://github.com/gioguerriero/ESA_Comet_Interceptor_Trajectory_Design_and_Optimization_Framework.git

global search with local refinement [16]. Grid-search sampling of the design space is often carried out beforehand to isolate promising regions and supply the initial guesses that improve convergence [17]. Together these tools provide the framework needed to study the escape dynamics from a Sun–Earth L_2 halo orbit and to design transfers exploiting both invariant-manifold structures and lunar gravity assists.

B. State of the Art

A comprehensive characterisation of the transfer problem has been presented by the ESA Mission Analysis Consolidated Report [18], which identifies two complementary departure strategies: a direct escape from L_2 and a Moon-gravity-assist escape, and maps their ΔV and time-of-flight performance across the full range of encounter geometries in the ecliptic plane within a simplified two-body Sun-centred dynamical model.

The heliocentric phase of the transfer has been further studied by Rebelo and Sánchez [19], who apply Primer Vector theory to optimise multi-DSM intercept trajectories for a set of recently discovered virtual targets, including a detailed launch-window analysis for comet C/2023 H2. That work, however, treats the Earth-escape phase in a patched-conic framework and does not model the three-body dynamics of the L_2 departure.

A dedicated analysis of this departure phase was carried out by Ranuschio [20], who adopts the Planar CR3BP and a Monte Carlo framework to demonstrate that optimal lunar flybys can be systematically targeted under the epistemic uncertainty on the relative spacecraft–Moon phasing at the time of comet discovery, confirming the statistical benefit of the swingby strategy in terms of both escape velocity and time of flight.

The geometry of the departure itself was subsequently studied in detail by Sánchez et al. [3], who show that the negative branch of the unstable manifold associated with the L_2 halo orbit naturally approaches the Earth–Moon system, and that a single lunar swingby can increase the v_∞ -equivalent escape velocity by approximately 200 m/s, providing a double benefit for Earth-leading intercept configurations: reduced ΔV cost and shortened time of flight.

The mechanism underlying the lunar encounter itself was studied by Chen et al. [21], who, working within a Sun–Earth CR3BP dynamical model, show that the first intersection of the Sun–Earth L_2 unstable manifold tube with the Moon’s orbit produces four distinct encounter geometries, each yielding a different v_∞ with respect to the Moon and hence a different post-swingby characteristic energy C_3 .

Taken together, these studies establish a solid dynamical characterisation of the departure problem, but remain primarily descriptive in nature: they map how the escape conditions vary with the halo geometry or the encounter type, without constructing, for a *given* target

comet, the actual transfer trajectory that exploits a lunar flyby while jointly minimising propellant cost and time of flight. Building on these foundations, the present work instead develops a systematic trajectory design and optimisation process: given the orbital state of a target comet, the proposed pipeline searches directly for transfer solutions that exploit a lunar flyby while optimising both the required ΔV and the time of flight, from halo-orbit departure to comet encounter within a single, three-dimensional dynamical model.

Two further key elements distinguish the present analysis from the body of work reviewed above. First, both the lunar-flyby and the direct-escape strategies are formulated and optimised within the *same* 3D Sun–Earth CR3BP framework, ensuring dynamical consistency across the halo departure, the lunar encounter, and the heliocentric leg. In this framework, the Moon is treated as a massless Keplerian body during the CR3BP propagation, meaning that its gravitational attraction is not explicitly included in the equations of motion. However, its effect is accounted for implicitly during the flyby through an instantaneous rotation of the spacecraft relative velocity vector, reproducing the gravity-assist effect of the lunar encounter. This dynamical framework is referred to hereafter as the *Sun–Earth CR3BP with Zero-SOI Moon* model. This represents an upgrade in fidelity over the simplified two-body dynamics model (Sun gravity only) used in [18] to generate the heat maps analyzing ΔV and ToF as functions of the target comet, as well as the patched-conic heliocentric optimisation of [19], and the planar CR3BP analysis of [20].

Second, both departure strategies are optimised on the *same* set of target comets: each of the seven comets analysed is used as the destination for both the lunar-flyby and the direct-escape strategy, so that their relative merits can be compared head-to-head on a per-target basis rather than across complementary regions of the encounter-geometry plane.

Specifically, the paper makes the following contributions:

1. A trajectory-optimisation pipeline is developed in the three-dimensional Sun–Earth CR3BP for the *lunar-flyby* departure strategy, encompassing the full sequence of halo-orbit departure, unstable-manifold coasting, instantaneous lunar swingby, and heliocentric transfer to the target comet, given the orbital state of a specified target comet.
2. An analogous optimised pipeline is constructed for the *direct-escape* strategy, in which the spacecraft is injected directly into the heliocentric leg without lunar encounter, ensuring a comparison under identical dynamical assumptions.
3. The two strategies are compared through ΔV –time-of-flight Pareto fronts computed for four representative target comets, chosen to sample the

four canonical encounter configurations: Earth-leading with $R_c < 1$ AU, Earth-leading with $R_c > 1$ AU, Earth-trailing with $R_c < 1$ AU, and Earth-trailing with $R_c > 1$ AU [18, 3]. The resulting fronts quantify, on a target-by-target basis, the conditions under which the lunar flyby provides a genuine advantage in propellant cost, transfer duration, or both.

4. The comparison is extended to three selected for an ESA Dry Run.*

The remainder of the paper is organised as follows. Section II describes the lunar-flyby pipeline, Section III the direct-escape one. Both are compared through ΔV –ToF Pareto fronts for four comets in Section IV and three Dry-Run targets in Section V. Sections VI and VII cover the ephemeris refinement and the energy analysis; Section VIII concludes.

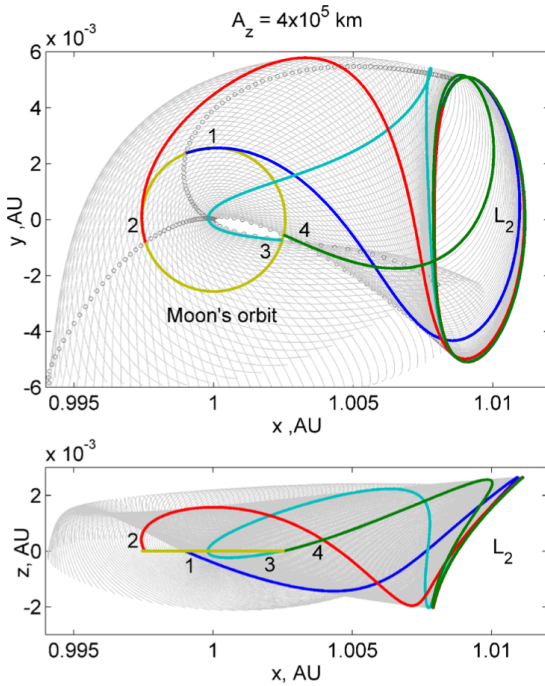


Fig. 1. Earth-bound branch of the Sun–Earth L_2 halo orbit unstable manifold intersecting lunar orbit [21].

II. Optimisation Process - Moon Flyby

This section presents the optimisation pipeline built for the lunar-flyby departure strategy: from the generation of the parking halo orbit and its unstable manifold to the multiple-shooting formulation of the lunar swingby and the heliocentric transfer toward the target comet.

A. Halo Orbit Generation and Validation

The departure of the Comet Interceptor spacecraft is set in a quasi-halo orbit around the Sun–Earth L_2 point.

*The ESA Dry Run was a mission-simulation exercise in which the full Comet Interceptor team rehearsed, end to end, the operational workflow to be followed after launch, in order to test how the decision-making process unfolds when selecting which of several candidate comets to target.

Halo orbits are three-dimensional periodic solutions of the Circular Restricted Three-Body Problem (CR3BP) that emerge from the linearised dynamics around the collinear libration points and extend into the nonlinear regime [5]. The instability of these orbits gives rise to invariant hyperbolic manifolds: a small perturbation along the unstable eigendirection naturally drives the spacecraft away from the periodic orbit along trajectories that, for the Sun–Earth L_2 case, can either depart outwards toward heliocentric space or fall inwards toward the Earth–Moon neighbourhood [21]. Figure 1 (from [21]) shows the Earth-bound branch of the unstable manifold tube, which intersects the Moon’s orbit, indicating that the natural dynamics already provide a corridor toward the lunar vicinity. This observation motivates the present work: the spacecraft can exploit the manifold flow to encounter the Moon at low cost, and use the resulting lunar flyby as a free energy source to improve the heliocentric leg toward the target comet in both ΔV and ToF.

Before deploying the full optimisation pipeline, the CR3BP implementation adopted in this work was validated against the reference results of Ranuschio [20] and Chen et al. [21] through two preliminary studies: the reconstruction of the parking halo orbit and the characterisation of the ballistic hyperbolic excess velocity v_∞ delivered at the lunar encounter. Together they provide an independent check of the correctness of the implementation prior to the analyses presented in the remainder of the paper.

The parking halo orbit was first rebuilt in order to reproduce, as closely as possible, the one adopted in the previous Comet Interceptor departure studies. Starting from a planar Lyapunov orbit around L_2 , a family of three-dimensional halo orbits is generated through a numerical continuation procedure, from which a member matching the orbital energy and geometric dimensions reported in [20] is selected. The two orbits, compared in Table 1, are deliberately kept very close to one another: the objective at this stage is not to identify a new optimal parking orbit, but to start from conditions directly comparable with previous work, so that the continuation and the subsequent manifold analysis remain consistent with the reference case.

Table 1: Comparison between the parking halo orbit reconstructed in this work and the one considered by Ranuschio [20].

	This work	Ranuschio [20]
Period [days]	179	178.6
Jacobi constant C	3.000716	3.000867
$A_{y,\max}$ [km]	870 228	871 000
$A_{z,\max}$ [km]	639 009	611 000

The second preliminary study addresses the ballistic hyperbolic excess velocity at the Moon. Chen et

al. [21] characterise the ballistic lunar encounters resulting from the four types of manifold crossings identified in their work by reporting the corresponding v_∞ as a function of the halo orbit amplitude A_z , showing that the four crossing types collapse onto two distinct v_∞ curves (the four different crossing types are represented in Fig. 1). Propagating the unstable manifold of the reconstructed halo orbit and detecting the resulting lunar encounters, the v_∞ delivered at the Moon was computed directly from the simulations and found to exhibit two distinct peaks, located at approximately $v_\infty \approx 0.75$ km/s and $v_\infty \approx 1.5$ km/s. Since the reference curves of [21] are provided only up to a limited halo amplitude, their data points were interpolated and the trend continued up to the amplitude of the orbit adopted here ($A_z = 6.39 \times 10^5$ km, see Table 1), yielding $v_\infty \approx 0.7497$ km/s and $v_\infty \approx 1.4352$ km/s. These extended values match the two peaks obtained from the present simulations, providing a further, independent confirmation that the manifold propagation and the encounter-detection logic are correctly implemented.

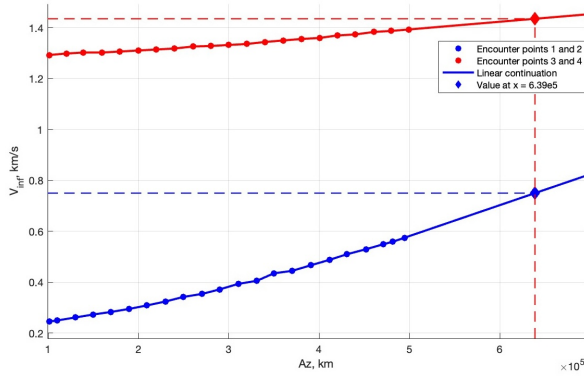


Fig. 2. Interpolated continuation of the ballistic lunar-encounter v_∞ as a function of the halo amplitude A_z . Chen et al.'s [21] reference curves (available only up to a limited A_z) are extrapolated to the halo amplitude adopted here, and the resulting two v_∞ peaks match those obtained from the present manifold simulations, validating the implementation.

B. Trajectory Architecture and Dynamical Model

The dynamical framework adopted throughout the optimisation is the three-dimensional Sun–Earth CR3BP, with the Moon included as a massless body on a circular, ecliptic-coplanar orbit and modelled with zero sphere of influence. The Moon's orbital eccentricity and inclination with respect to the ecliptic are therefore neglected to keep the model computationally light. Although a dedicated full-ephemeris refinement algorithm has been developed to validate the fidelity of this simplified model starting from CR3BP solutions, the high-fidelity propagation is not the subject of the present paper, and all the trajectories shown here are generated within the CR3BP framework.

Within this dynamical setting, the trajectory from the

halo orbit to the comet is decomposed into four arcs separated by three impulsive manoeuvres and one instantaneous lunar flyby:

- A small injection ΔV_{inj} at the halo departure point inserts the spacecraft into the unstable manifold, initiating the first ballistic arc.
- A first deep-space manoeuvre, ΔV_{DSM1} , is then applied to deflect the natural manifold trajectory toward the desired lunar encounter conditions.
- The lunar flyby is modelled, consistently with the zero-SOI assumption, as an instantaneous rotation of the spacecraft's hyperbolic excess velocity vector v_∞ relative to the Moon.
- After the flyby, a second arc carries the spacecraft into the heliocentric regime, where a second deep-space manoeuvre, DSM2, provides the final phasing correction needed to target the comet at the prescribed encounter epoch.

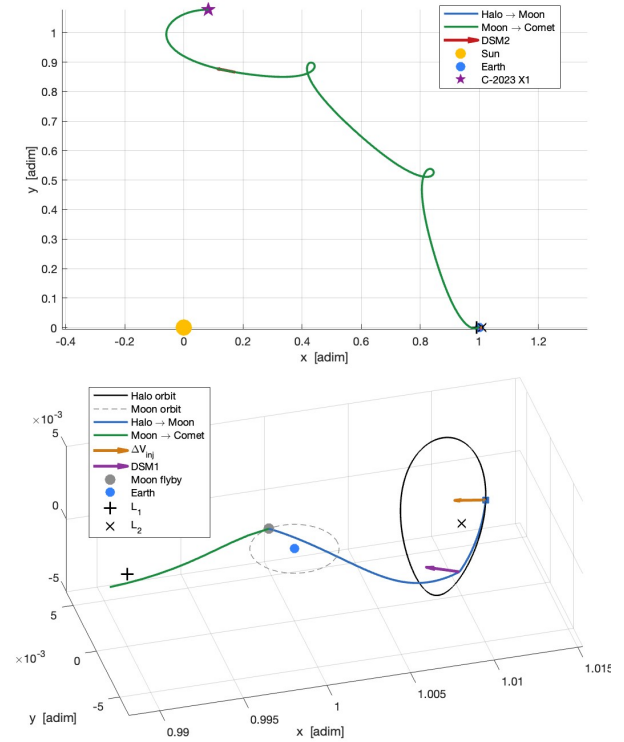


Fig. 3. Sample mission trajectory targeting comet C/2023 X1 in the Sun–Earth synodic frame (top) and zoomed view of the initial halo-orbit departure (bottom).

It is worth noting that, for the analysis of the selected target comets (that will be presented in Section IV), the target point was defined as the intersection between the comet trajectory and the Earth's ecliptic plane. This choice represents a very good approximation of the optimal intercept point, as shown in [19].

As an illustrative example, Figure 3 shows a sample trajectory targeting comet C/2023 X1, representative of the mission architecture described above.

C. Optimisation Pipeline

The trajectory design problem described above is solved by splitting the transfer into two consecutive optimisation phases, namely a *Moon-to-Comet* (M2C) phase and a *Halo-to-Moon* (H2M) phase.

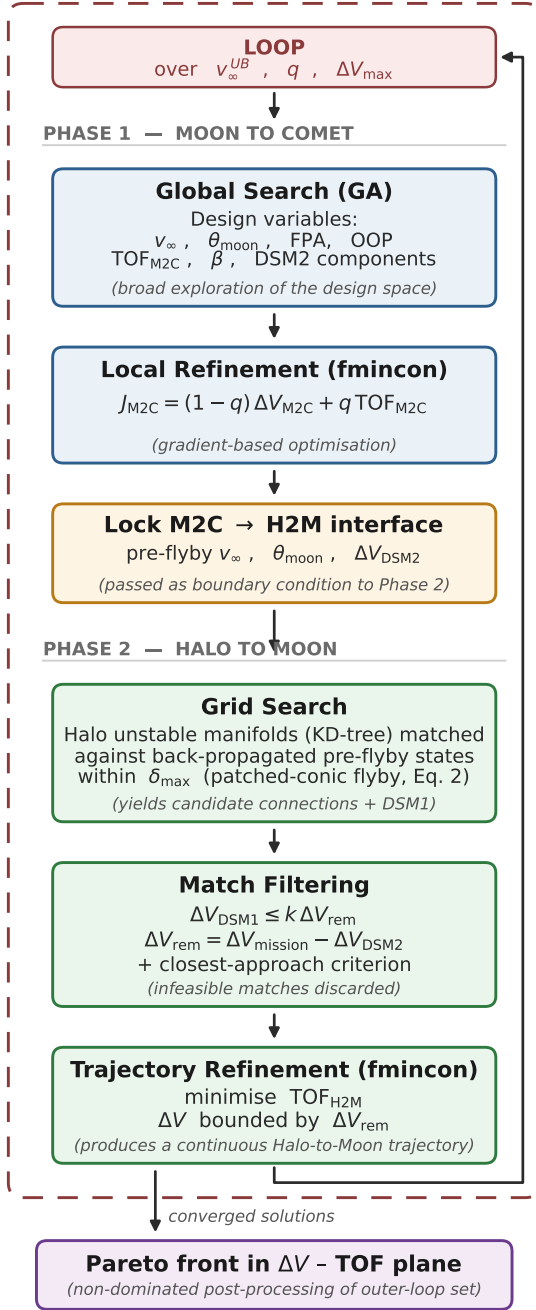


Fig. 4. Optimization pipeline flowchart.

The order is dictated by the nature of the boundary conditions: the comet imposes a rigid rendezvous in

position at a prescribed epoch, whereas the halo departure offers a continuous one-parameter family of natural manifold trajectories. In reality, the relative phase between the spacecraft along the halo orbit and the Moon at the time of comet discovery is fixed by the mission timeline and cannot be chosen freely; in the present work, however, this phase is treated as a free variable.

The M2C phase is therefore optimised first, and the resulting lunar encounter state is then used as a target for the H2M phase. A schematic summary of the process is shown in Figure 4.

The M2C phase is parametrised by the following design variables: the lunar excess velocity magnitude v_∞ , the lunar phase angle θ_{moon} , the flight-path angle (FPA) and out-of-plane angle (OOP) of the post-flyby v_∞ vector, the time of flight ToF_{M2C} , the split fraction β that locates DSM2 along the heliocentric leg, and the three components of the DSM2 impulse.

The cost function combines propellant and time through the weighted form

$$J_{\text{M2C}} = (1 - q) \cdot \Delta V_{\text{M2C}} + q \cdot \text{ToF}_{\text{M2C}}, \quad (1)$$

where $q \in [0, 1]$ is a scalar weight, looped externally in the nested loop described below, that balances the relative importance of ΔV and ToF, in order to generate a discretized set of solutions within the Pareto Front. The optimisation is performed in two steps: a global exploration with a genetic algorithm provides an initial guess that is subsequently refined with a gradient-based solver (fmincon), the algorithmic details being beyond the scope of this paper.

Once the M2C phase has converged, the optimised lunar encounter state (specifically the pre-flyby v_∞ vector and the lunar phase θ_{moon}) is passed to the H2M phase as a boundary condition, together with the value of ΔV_{DSM2} . The H2M phase is initialised through a manifold-matching procedure designed to identify dynamically natural connections between the halo orbit and the prescribed lunar encounter condition.

First, a database of unstable manifold trajectories emanating from the halo orbit is generated once by perturbing (with a $\Delta V_{\text{inj}} = 15$ m/s) a discretised set of halo departure points along the unstable eigendirection and propagating forward in the CR3BP. All sampled points along these trajectories are stored in a KD-tree, indexed by trajectory and time, which is then reused across all subsequent optimisation runs to reduce computational cost.

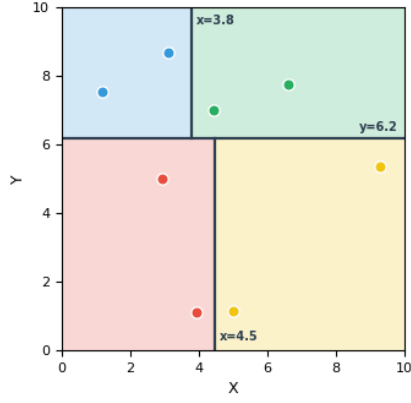


Fig. 5. K-d tree construction on a 2D example with $N = 8$.

The unstable manifold is propagated from 3500 equally-spaced points on the Halo orbit, yielding a database of approximately $N = 5\text{--}7 \times 10^6$ three-dimensional position vectors in the CR3BP synodic frame. These points are organized into a KD-tree structure, which recursively partitions the point cloud by splitting at each level along the dimension of maximum variance and dividing at the median. With a leaf size of 50, the resulting tree has approximately 17 levels. Candidate connection points are retrieved via a range-search query with a threshold of 20,000 km, reducing the nearest-neighbor search from $\mathcal{O}(n)$ to $\mathcal{O}(\log n)$ and making the matching step computationally tractable.

To clarify the partitioning logic used in this work, Figure 5 shows a simplified two-dimensional example with $N = 8$ points: at each level, the current point set is split at the median along one coordinate axis, producing the four rectangular leaf cells shown in the scatter plot above and the corresponding binary tree below. Note that MATLAB's `KDTreeSearcher` function was used for this work.

In parallel, the target pre-flyby state is back-propagated in the CR3BP, and a discrete set of admissible pre-flyby states is generated by sampling the rotations of the v_∞ vector within the maximum bending angle allowed by a patched-conic flyby. The maximum

bending angle $\delta_{\max}$ is computed as (from [13]):

$$\delta_{\max} = 2 \arcsin \left(\frac{\mu_M}{v_\infty^2 \cdot (R_M + h_p) + \mu_M} \right) \quad (2)$$

where v_∞ is the magnitude of the hyperbolic excess velocity at infinity, R_M is the Moon's radius, h_p is the flyby altitude, and μ_M is the Moon's gravitational parameter. Note that throughout this work a flyby altitude of $h_p = 750 \text{ km}$ has been adopted, consistent with the flyby altitude demonstrated by the JUICE mission [22].

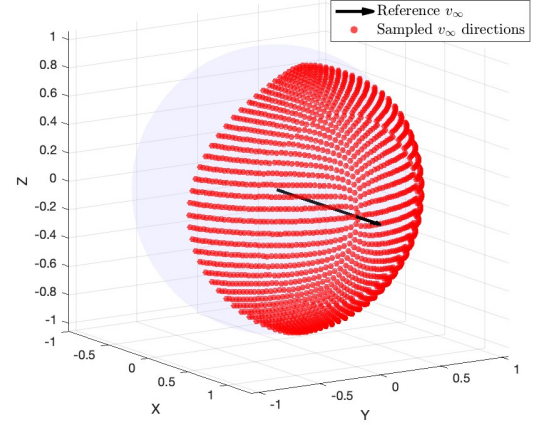


Fig. 6. 2000 sampled v_∞ directions (red) for $\delta_{\max} = 70^\circ$; the black arrow is the reference v_∞ .

The admissible v_∞ directions are sampled on the deflection cone of half-aperture $\delta_{\max}$ using a polar grid with uniform areal density: concentric rings of equispaced radii, with the number of angular samples on each ring proportional to its radius. This yields a more uniform distribution of points over the spherical cap than a rectangular grid in the deflection angles. A total of $M = 2000$ directions are sampled; for each, the rotated v_∞ is added to the Moon's velocity and back-propagated in the CR3BP to build the database of admissible pre-flyby trajectories. Fig. 6 shows the resulting distribution of the 2000 sampled v_∞ directions for a maximum bending angle of 70° .

The set of admissible back-propagated trajectories is then queried against the halo KD-tree to identify spatial matches, each of which provides a candidate Halo-to-Moon connection together with the velocity mismatch that has to be covered by the first deep-space manoeuvre, DSM1.

The raw set of matches is filtered through two criteria. The first is a ΔV feasibility filter: each candidate DSM1 must satisfy

$$\begin{aligned} \Delta V_{\text{DSM1}} &\leq k \cdot \Delta V_{\text{rem}}, \\ \Delta V_{\text{rem}} &= \Delta V_{\text{max}} - \Delta V_{\text{DSM2}} - \Delta V_{\text{inj}}, \end{aligned} \quad (3)$$

where ΔV_{max} is the maximum mission budget, ΔV_{inj} is the small injection maneuver to leave the Halo ($\Delta V_{\text{inj}} =$

15 m/s for this work) and k is a tuning factor that controls how much of the remaining budget (ΔV_{rem}) is allowed in the matching phase ($k = 3$ for this work). The second is a spatial filter based on the closest point of approach between the forward halo trajectory and the back-propagated Moon trajectory, which discards matches whose geometric mismatch is too large to be reconciled by a small correction (mismatch $< 20,000$ km for this work). The surviving candidates are then refined individually: each match is used as an initial guess for a local fmincon optimisation that produces a continuous Halo-to-Moon trajectory. In the refinement, only the time of flight is minimised, while the total ΔV is enforced as a hard constraint bounded by ΔV_{rem} .

The procedure described so far produces a single Halo-to-Moon to-Comet trajectory for a given set of bounding parameters. To obtain a complete Pareto front in the $\Delta V - \text{ToF}$ plane, the entire pipeline is embedded in a triple nested loop over:

- the lunar flyby excess velocity upper bound v_{∞}^{UB} , which controls the energy regime of the flyby;
- the cost-function scalar weight q in Eq. (1);
- the maximum mission ΔV (ΔV_{max}) which is varied to populate the Pareto front along the ΔV axis.

The full set of converged solutions across the outer sweep is post-processed to extract the non-dominated front in the $\Delta V - \text{ToF}$ plane, which constitutes the primary output of the methodology and the object that will be compared, target by target, with the corresponding front obtained from the direct-escape strategy.

III. Optimization process - Direct Injection

The optimization process for the direct injection strategy, which does not include a lunar flyby, is described in this section.

For this strategy, the trajectory departing from the halo orbit is split into two arcs. The first arc is the exit of Comet-Interceptor from the halo orbit along an unstable manifold. The second arc is a Lambert arc obtained by solving a two-point boundary value problem in the CR3BP. The initial halo orbit is fixed, moreover the interception date and thus the coordinate of the intercept point is pre-optimized in the two body problem. The problem is then formulated using 4 variables :

- m : departing phase on the halo orbit. m is normalized between 0 and 1.
- ToF : total time of flight.
- ΔV_1 : amplitude of the first impulse. The direction of the maneuver being fixed to the unstable invariant manifold.

- β : ratio between t_1 and ToF , where t_1 is the time spent on the manifold arc.

The cost function is the ΔV needed for the trajectory :

$$J = ||(\Delta V_1)|| + ||(\Delta V_2)||$$

where $||(\Delta V_1)||$ is a variable and $||(\Delta V_2)||$ is calculated via the Lambert arc.

In order to robustly explore the search space, a global, derivative-free optimization method based on Differential Evolution (DE) is adopted using the implementation provided by the Scipy library. Once a promising solution is identified, it is subsequently refined using a gradient-based L-BFGS solver to improve local accuracy. DE is particularly suited to this problem because it can enable a large exploration of the search space and handles highly nonlinear, nonconvex and multimodal objective functions.

IV. Results and analysis

A. Selected Target Comets

As anticipated in the introduction, four representative target comets were selected to sample the canonical encounter-geometry space defined by two parameters: the leading or trailing configuration of the comet with respect to the Earth in the synodic frame, and the heliocentric distance at encounter relative to 1 AU (Fig. 7). The four target comets and their encounter-geometry parameters are summarized in Table 2, where the angular separation is also reported for reference.

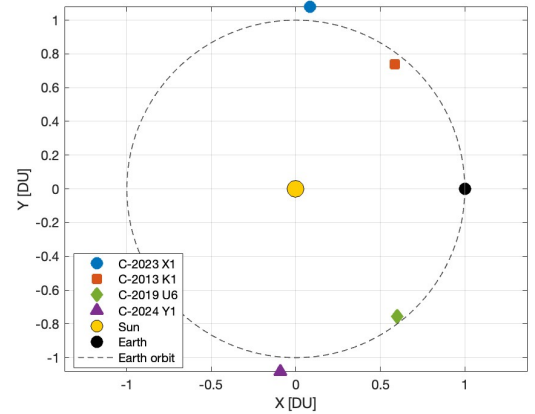


Fig. 7. Selected target comets in the Sun–Earth synodic frame.

Table 2: Selected target comets and their encounter-geometry parameters.

Comet	Configuration	R_c	Angular sep.
C/2023 X1	Leading	> 1 AU	Large
C/2013 K1	Leading	< 1 AU	Small
C/2019 U6	Trailing	< 1 AU	Small
C/2024 Y1	Trailing	> 1 AU	Large

This selection ensures that the Pareto-front comparison between the lunar-flyby and direct-escape strategies covers a representative portion of the encounter-geometry plane. It is worth noting that the choice to avoid heliocentric distances significantly far from 1 AU and angular separations substantially larger than those listed is driven by mission feasibility: comets with such extreme geometries would require propellant budgets exceeding the ΔV allocation available to the Comet Interceptor mission.

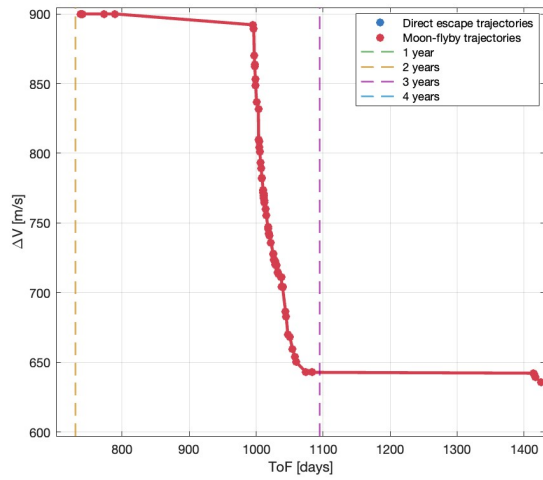
B. Pareto Front Analysis

The ΔV -ToF Pareto fronts for the four selected target comets are presented in Fig. 8, where each panel compares the lunar-flyby strategy with the direct-escape

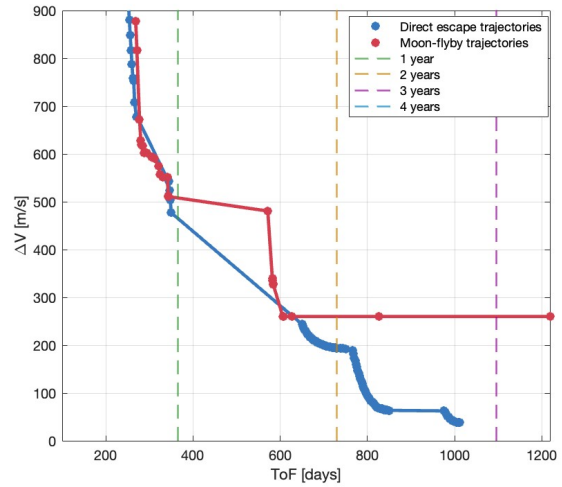
baseline.

1) C/2023 X1

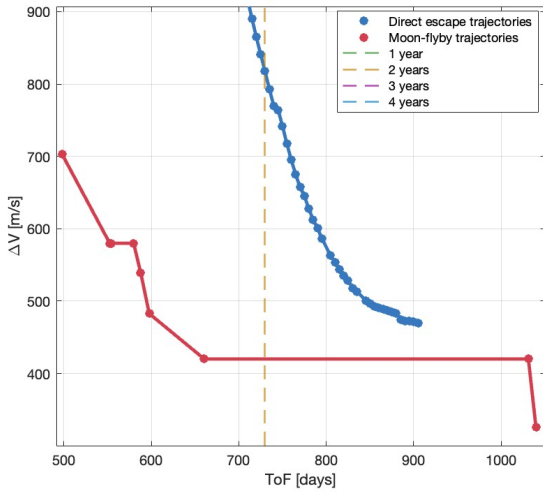
C/2023 X1, the Earth-leading comet at $R_c > 1$ AU, represents the most demanding intercept configuration in the selected set. The direct-escape Pareto front lies entirely outside the accessible mission envelope: no trajectory satisfying the $\Delta V_{\max} = 900$ m/s budget can reach this target without a lunar flyby. The swingby is therefore not merely advantageous but strictly necessary, providing the additional characteristic energy required to bridge the gap between the L2 departure v_∞ and the heliocentric injection requirement imposed by this geometry.



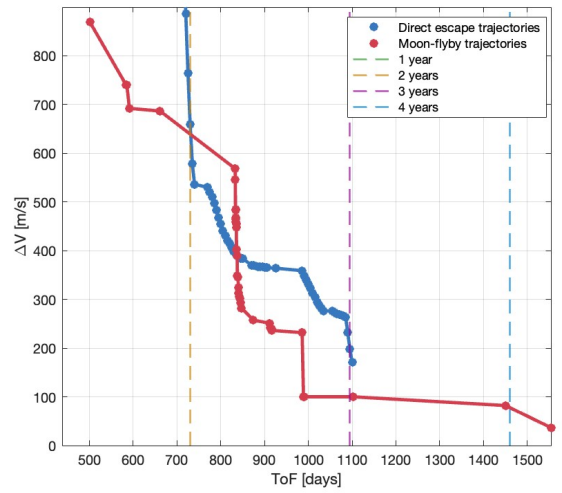
C/2023 X1



C/2013 K1



C/2019 U6



C/2024 Y1

Fig. 8. ΔV -ToF Pareto fronts for the four selected target comets. Blue: direct escape trajectories. Red: lunar-flyby trajectories.

2) C/2013 K1

For C/2013 K1, an Earth-leading comet at $R_c < 1$ AU with small angular separation, the picture reverses completely. The lunar-flyby Pareto front either overlaps with or is outperformed by the direct-escape front across the ΔV budget range. A notable consequence is the appearance of a hard lower bound on the achievable budget: the flyby strategy cannot deliver the spacecraft to the comet below approximately 260 m/s, a threshold that the direct strategy undercuts. This behaviour is consistent with the geometric argument that a nearby, nearly in-plane target imposes mild energy requirements already satisfied by the direct departure; the additional arc needed to target the Moon then introduces a net ΔV penalty with no compensating gain in escape velocity. For this configuration, the operational overhead of a lunar flyby is not justified by any performance gain.

3) C/2019 U6

C/2019 U6, an Earth-trailing comet at $R_c < 1$ AU, exhibits a clear benefit when using a lunar flyby to target it. As can be seen in the bottom-left plot, the flyby front lies consistently below the direct-escape front: at any fixed ΔV level the swingby delivers a shorter transfer time. More significantly, the flyby extends the accessible budget range toward lower values, reducing the minimum achievable ΔV from approximately 470 m/s under direct escape to roughly 325 m/s, a reduction of nearly 30%. This extension opens a low-cost intercept corridor that is entirely inaccessible to the direct strategy.

4) C/2024 Y1

The trailing comet at $R_c > 1$ AU, C/2024 Y1, presents the most nuanced comparison of the set, with three distinct regimes identifiable along the Pareto front. At high ΔV , the flyby strategy achieves shorter transfer times, making it attractive for time-critical scenarios with ample propellant budget. At intermediate ΔV , the direct-escape front dominates, with the lunar detour introducing a net time penalty. At low ΔV , the flyby recovers its advantage and extends the reachable budget range far more significantly than in the C/2019 U6 case: the minimum accessible ΔV drops from approximately 170 m/s under direct escape to roughly 36 m/s under the flyby strategy. This near-total elimination of the propellant requirement implies that, for certain departure epochs, the lunar gravity assist alone is nearly sufficient to inject the spacecraft onto an intercept trajectory, with the remaining impulsive manoeuvres acting only as small corrective impulses.

V. Extended Case Study: ESA Dry-Run Targets

A. Selected Targets

The four-comet analysis presented in Section IV

was conceived to span the canonical encounter geometry space in a systematic way. To further stress-test the pipeline in an operationally realistic setting, the case study is here extended to three additional targets selected by ESA for a *Dry Run* of the mission decision process. The Dry Run is designed to reproduce the moment in which a suitable comet is identified and a departure strategy must be committed to, while the spacecraft is already in its parking orbit around the Sun–Earth L_2 point.

The three Dry-Run comets, referred to as Comet 1, Comet 2 and Comet 3, all share the same broad encounter geometry, being Earth-trailing targets at $R_c > 1$ AU, but differ in their angular separation from the Earth in the synodic frame as well as the distance from the Sun. Their encounter geometry parameters are summarised in Table 3, and their positions in the Sun–Earth synodic frame are shown in Fig. 9.

Table 3: ESA Dry-Run target comets and their encounter-geometry parameters.

Comet	Configuration	R_c	Angular sep.
Comet 1	Trailing	> 1 AU	Large
Comet 2	Trailing	> 1 AU	Small
Comet 3	Trailing	> 1 AU	Very small

B. Pareto Front Analysis

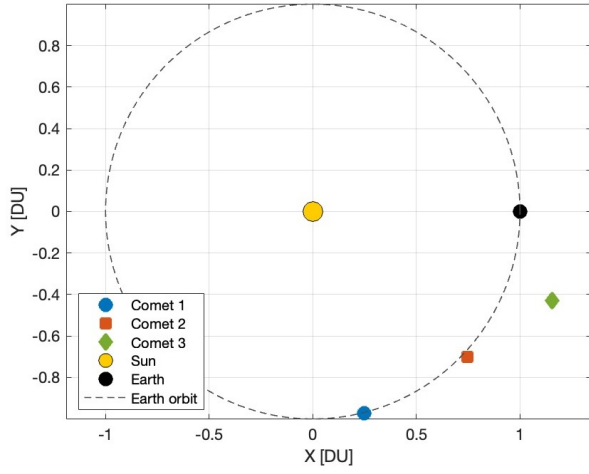
The ΔV –ToF Pareto fronts for the three Dry-Run comets are presented in Fig. 9, following the same convention adopted in Section IV: each panel compares the lunar-flyby strategy (red) with the direct-escape baseline (blue).

1) Comet 1

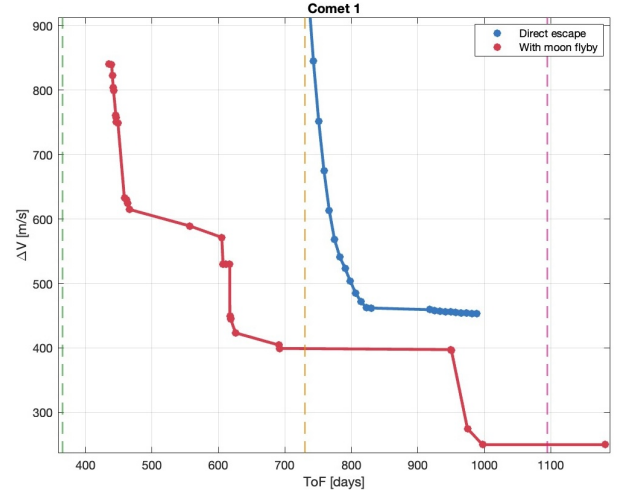
This comet exhibits a clear and consistent benefit from the lunar flyby, with a behaviour similar to C/2019 U6. Across the entire ΔV budget range, the swingby front lies below the direct-escape front, delivering a transfer-time saving of at least 150 days at any fixed ΔV level. In addition, the flyby strategy extends the accessible envelope toward lower ΔV values, opening the possibility of reaching the comet at reduced propellant cost at the expense of longer times of flight. For this geometry the lunar assist is therefore advantageous.

2) Comet 2

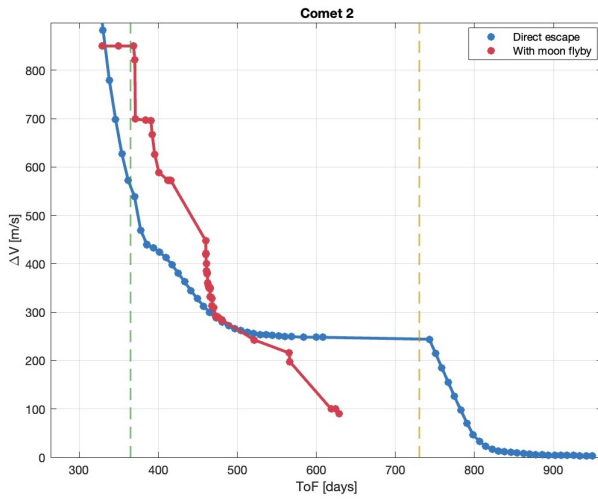
This comet, characterised by a small angular separation, reverses the picture entirely, as in the C/2013 K1 case. The lunar-flyby front provides no benefit over the direct-escape strategy, the additional arc required to target the Moon introducing a net penalty with no compensating gain in escape velocity. Remarkably, the direct-escape strategy admits a nearly cost-free transfer provided a sufficiently long waiting time is accepted: the minimum achievable ΔV collapses to values close to zero.



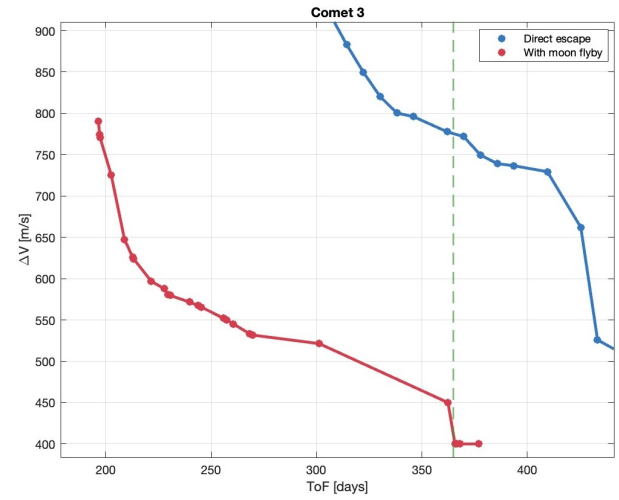
Target comets



Comet 1



Comet 2



Comet 3

Fig. 9. ESA Dry-Run target comets in the Sun–Earth synodic frame (top left), and ΔV –ToF Pareto fronts for the three Dry-Run targets (top right, bottom left, bottom right). Blue: direct-escape trajectories. Red: lunar-flyby trajectories.

This behaviour is a direct consequence of the encounter geometry: as will be discussed in the Conclusions, Comet 2 lies almost exactly at 1 AU, in the middle of the corridor swept by the unstable manifolds emanating from the L_2 halo orbit, so that the natural dynamics alone are sufficient to reach it. For this target the operational overhead of a lunar flyby is not justified.

3) Comet 3

This is the target with the smallest angular separation, nonetheless recovers a marked benefit from the lunar flyby. As for Comet 1, the swingby front provides a transfer-time reduction of at least 150 days at any fixed ΔV with respect to the direct-escape baseline.

D. Common Optimal Geometry: Powered Earth Flyby

A noteworthy feature emerges from the optimisation of these three Dry-Run targets: the best-performing solutions are all realised through the *same* trajectory geom-

etry. In each case, the optimiser exploits the lunar flyby not to inject the spacecraft directly onto the heliocentric leg, but to steer it toward a subsequent *low Earth flyby*, at which a large impulsive manoeuvre is applied. Performing the manoeuvre deep inside the Earth's gravity well, where the spacecraft velocity is at its highest, maximises the resulting energy gain through the Oberth effect. This is made explicit by the energy increment associated with an impulsive manoeuvre of magnitude $\Delta \mathbf{v}$ applied when the spacecraft has velocity $\mathbf{V}$:

$$\delta \varepsilon = \frac{|\mathbf{V} + \Delta \mathbf{v}|^2}{2} - \frac{|\mathbf{V}|^2}{2} = \mathbf{V} \cdot \Delta \mathbf{v} + \frac{1}{2} |\Delta \mathbf{v}|^2 \quad (4)$$

The dominant term $\mathbf{V} \cdot \Delta \mathbf{v}$ scales linearly with the local speed $|\mathbf{V}|$: for a fixed manoeuvre magnitude, the same ΔV therefore causes a much larger increase in orbital energy when it is spent at high speed, i.e. at the perigee of a close Earth flyby, than when it is spent slowly far from any massive body. The lunar flyby thus acts as a

phasing enabler that positions the spacecraft for a powered Earth flyby, and it is the combination of the two encounters that produces the fastest escape trajectories (lowest Time of Flight) for the Dry-Run targets.

A representative example of this optimal geometry is shown in Fig. 10.

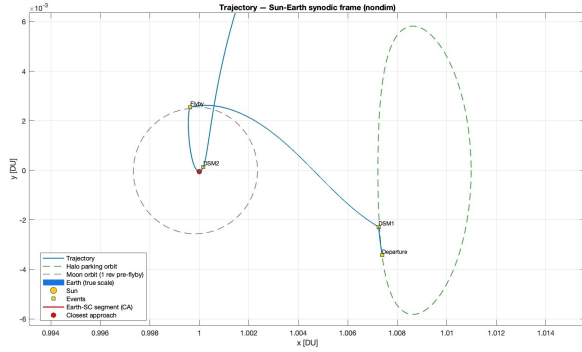


Fig. 10. Sample trajectory realising the common optimal geometry for the Dry-Run targets: lunar flyby followed by a powered low Earth flyby, shown in the Sun–Earth synodic frame (targeting Comet 3 in this specific case).

VI. Ephemeris Refinement

The trajectories presented throughout this work are computed within the simplified Sun–Earth CR3BP with a massless, zero-SOI Moon. To provide evidence that at least a subset of these solutions survives in a realistic dynamical environment while retaining approximately the same ΔV and time of flight, a refinement procedure toward a full-ephemeris model was carried out. The refinement is organised in two successive steps.

The first step transitions from the CR3BP to a model in which the Earth and Moon follow their real ephemeris positions, while the gravitational attraction of only the Sun and the Earth is retained; the Moon is still treated with a zero sphere of influence, its effect being reproduced by the instantaneous v_∞ rotation as in the CR3BP formulation. This intermediate model proved robust: solving the associated boundary-value problem by multiple shooting converged in practically all cases, at most requiring a modest adjustment of the number of nodes allocated to each phase. The plot in Fig. 10 represents a trajectory in this dynamical framework.

The second step transitions to the fully realistic model, in which the Moon’s gravity is also explicitly included. To provide the additional degrees of freedom needed to absorb the perturbation introduced by the lunar attraction, two trajectory correction manoeuvres (TCMs) were added a few days before and a few days after the lunar flyby. For a number of solutions this procedure yielded trajectories fully propagated in the real-ephemeris model, with ΔV and time-of-flight values consistent with those of the original CR3BP solutions, thereby confirming that the latter do exist in a real-

istic dynamical setting. For other solutions, however, the procedure encountered convergence difficulties, and the method could therefore not be considered fully robust.

For these reasons, the complete refinement algorithm is not presented here; it is reported only that the trajectories were successfully refined into a real-ephemeris model, although not for all cases and not in a fully systematic and direct manner. A robust, general-purpose refinement pipeline is left to future work, possibly building on the ESA GODOT library, whose integration was initiated in the course of this study but could not be completed within the available time.

As an illustrative example, Fig. 11 compares, for comet C/2019 U6, the Pareto front obtained from the CR3BP global search with the Pareto front formed by the subset of those solutions successfully refined into the full-ephemeris model (Sun–Earth–Moon). The comparison shows that the refiner does not manage to converge for every solution on the original front, consistently with the lack of robustness discussed above. It is also apparent that the time of flight of the refined solutions remains essentially unchanged with respect to their CR3BP counterpart (as expected, since the ToF is enforced as a constraint of the optimisation process) whereas the corresponding ΔV is considerably more variable and not robust, in some cases increasing and in others decreasing with respect to the original CR3BP value. This behaviour is attributed to the refinement algorithm itself, rather than to the dynamical-model transition, which is carried out smoothly and without any aggressive jump between models (CR3BP $\rightarrow$ Sun–Earth ephemeris $\rightarrow$ Sun–Earth–Moon ephemeris); the most likely cause is instead believed to lie in the way the refinement procedure was implemented. Fixing this issue is therefore expected to be a promising and rewarding direction for future work.

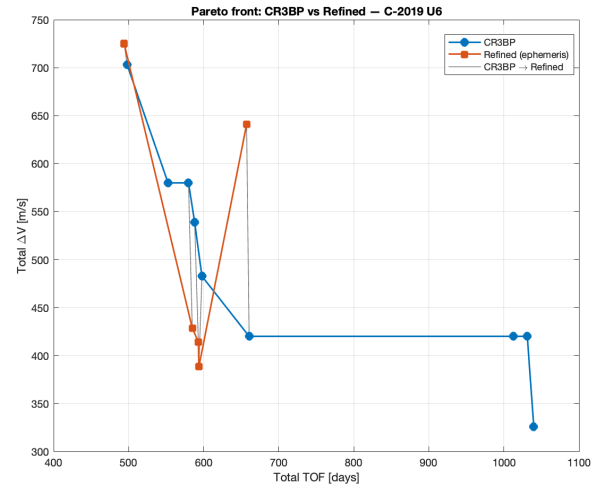
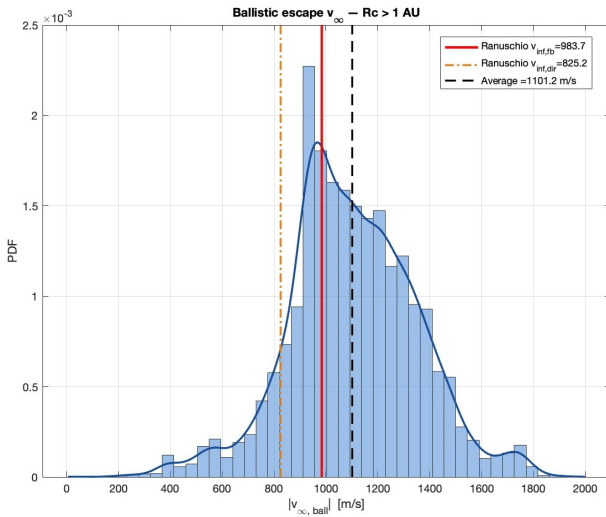
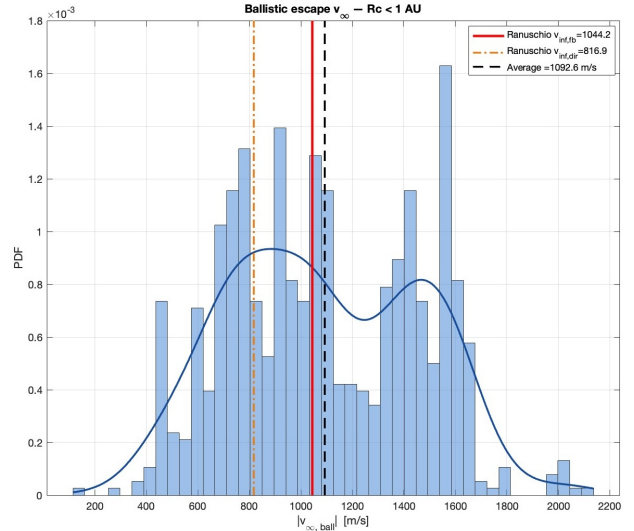


Fig. 11. Comparison, for comet C/2019 U6, between the Pareto front from the CR3BP global search and the Pareto front of the solutions refined in the full-ephemeris (Sun–Earth–Moon) model.



$R_c > 1$ AU (~ 5500 solutions)



$R_c < 1$ AU (~ 800 solutions)

Fig. 12. Distribution of the ballistic-equivalent escape velocity $|v_{\infty,ball}|$ for the two heliocentric-distance classes. Red: reference flyby value from [20]. Orange dash-dotted: reference direct-escape value. Black dashed: mean of the present distribution.

VII. Energy-Based Analysis of the Ballistic Escape Velocity

A. Motivation and Method

A complementary way of assessing the benefit of the lunar swingby, and of validating the present results against previous work, is to look at the Earth-escape velocity itself rather than at the mission-level ΔV -ToF trade-off. This is the approach followed by Ranuschio [20], who evaluated the heliocentric escape v_{∞} at a fixed control radius r_c around the Earth and found that the lunar flyby raises it to approximately 1 km/s, against roughly 0.82 km/s for the direct escape, i.e. a free gain of about 200 m/s.

A direct comparison with those figures is not immediate, however. The reference results are obtained for *purely ballistic* escape trajectories, whereas the solutions produced by the present pipeline generally contain impulsive manoeuvres (halo injection, DSM1 and, in some cases, DSM2) that are executed *inside* the control radius and therefore contribute part of the escape energy.

To enable a fair comparison, the energy imparted by these manoeuvres is removed from the final state, accounting for the point along the trajectory at which each manoeuvre is applied.

The procedure is as follows. The heliocentric specific orbital energy is first evaluated at the outward crossing of the control radius, here set to $r_c = 0.2$ AU consistently with [20]:

$$\varepsilon_f = \frac{V^2}{2} - \frac{\mu_{\odot}}{r}. \quad (5)$$

The energy increment produced by each manoeuvre executed before the control radius is crossed is then com-

puted through the same relation already introduced in Eq. (4), $\delta\varepsilon_i = \mathbf{V}_i \cdot \Delta\mathbf{v}_i + \frac{1}{2}|\Delta\mathbf{v}_i|^2$, where $\mathbf{V}_i$ is the heliocentric velocity at the manoeuvre point. Evaluating $\delta\varepsilon_i$ with the local velocity is essential: the same ΔV contributes a different amount of energy depending on where along the trajectory it is spent. The propulsive contribution is then subtracted from the final energy,

$$\varepsilon_{ball} = \varepsilon_f - \sum_i \delta\varepsilon_i, \quad (6)$$

yielding the osculating semimajor axis of the equivalent ballistic heliocentric orbit,

$$a_{ball} = -\frac{\mu_{\odot}}{2\varepsilon_{ball}}, \quad (7)$$

and finally the corresponding escape velocity, defined as the difference between the velocity on the equivalent orbit at the Earth's heliocentric distance r_E and the Earth's circular velocity:

$$v_{\infty,ball} = \sqrt{2\left(\frac{\mu_{\odot}}{r_E} - \frac{\mu_{\odot}}{2a_{ball}}\right)} - \sqrt{\frac{\mu_{\odot}}{r_E}}. \quad (8)$$

Applying Eqs. (5)–(8) to the full set of converged lunar flyby solutions produces, for each geometry class, a distribution of $v_{\infty,ball}$ that can be compared directly with the reference values of [20].

B. Results and Discussion

The two distributions are shown in Fig. 12. The mean ballistic-equivalent escape velocity obtained here is 1101.2 m/s for the $R_c > 1$ AU class and 1092.6 m/s for the $R_c < 1$ AU class, against the reference flyby values of 983.7 m/s and 1044.2 m/s respectively. The agreement of the mean values is the main outcome of this analysis: once the propulsive contribution is stripped

away, the escape energy delivered by the lunar encounter in the present three-dimensional framework is consistent with the independent estimate of [20], and comfortably above the corresponding direct-escape references (825.2 m/s and 816.9 m/s). This provides an independent, energy-level validation of the swingby benefit reported in earlier studies.

Three observations qualify this result. First, the two distributions are not equally well resolved: the $R_c > 1$ AU class is supported by roughly 5500 converged solutions against approximately 800 for the $R_c < 1$ AU class, simply because five of the seven comets studied in this work lie at $R_c > 1$ AU. The smoother, clearly unimodal shape of the former is therefore at least partly a sampling effect, and the pronounced multi-modality of the latter may well be an artefact of the limited sample rather than an actual dynamical feature.

Second, and more fundamentally, the optimisation performed here does not maximise v_∞ , whereas the reference study does: the present cost function targets ΔV and time of flight toward a specific comet, so that the escape velocity is an outcome rather than an objective. This explains the non-negligible fraction of solutions whose $v_{\infty, \text{ball}}$ falls even below the direct-escape reference. Such solutions are not inefficient: a lower escape velocity pointed in a more favourable direction can reach the target comet more cheaply than a larger but poorly oriented one. The magnitude of v_∞ alone is therefore an incomplete figure of merit in the present context, which is precisely why the ΔV –ToF Pareto fronts, and not the escape velocity, are adopted as the primary metric of this work.

Third, the procedure carries an intrinsic conceptual limitation. Removing the energy of the manoeuvres does not remove their *geometric* effect: it is precisely thanks to those manoeuvres that trajectory shapes are accessed which would simply not exist in a purely ballistic flow. The quantity $v_{\infty, \text{ball}}$ should therefore be read as a ballistic-equivalent escape velocity, useful for a comparison at the energy level, rather than as the escape velocity of an actual ballistic trajectory.

VIII. Conclusions

This work has investigated the benefit of incorporating a single lunar gravity assist into the Earth-escape segment of the ESA Comet Interceptor mission, by formulating both the lunar-flyby and the direct-escape strategies within a common three-dimensional Sun–Earth CR3BP framework and comparing them, on a per-target basis, through ΔV –ToF Pareto fronts. Four representative comets were first selected to sample the canonical encounter-geometry configurations (Earth-leading/trailing, $R_c > 1$ AU and $R_c < 1$ AU), and the analysis was then extended to three additional Earth-trailing, $R_c > 1$ AU targets selected for an ESA Dry Run, for a total of seven comets assessed under identical

dynamical assumptions and identical target sets.

The analysis shows that the merit of the lunar swingby is strongly geometry-dependent, and that its advantage is most pronounced precisely in those regions of the encounter-geometry plane that are not naturally reached by the unstable manifold emanating from the Sun–Earth L_2 halo orbit.

As illustrated in Figure 12, the unstable manifold, viewed from above the ecliptic, naturally flows toward Earth-leading configurations at $R_c < 1$ AU and Earth-trailing configurations at $R_c > 1$ AU, sweeping past comets such as C/2013 K1 and C/2024 Y1 respectively; for these targets, the direct-escape strategy is already well-supported by the natural dynamics, and the operational overhead of a lunar encounter is either marginal or not justified.

In contrast, for Earth-leading targets at $R_c > 1$ AU and Earth-trailing targets at $R_c < 1$ AU, the manifold does not naturally provide the energy or the geometry required, and the lunar flyby becomes either genuinely advantageous (as for C/2019 U6, where it reduces the minimum required ΔV by nearly 30%) or strictly necessary (as for C/2023 X1, where no direct-escape trajectory exists within the mission budget).

The three ESA Dry-Run targets refine, rather than contradict, this picture. Although all three share the same geometry (Earth-trailing, $R_c > 1$ AU), they do not behave uniformly, which shows that the leading/trailing and $R_c \lesseqgtr 1$ AU classification is only a first-order proxy. The truly discriminating criterion is whether the point at which the comet crosses the ecliptic plane coincides with the region actually swept by the unstable manifolds. Comet 2 crosses the ecliptic almost exactly at 1 AU, right in the middle of the manifold corridor, so that the natural dynamics alone are essentially sufficient to reach it and the lunar flyby brings no benefit at all; the direct-escape strategy even admits a nearly cost-free transfer for a sufficiently long waiting time.

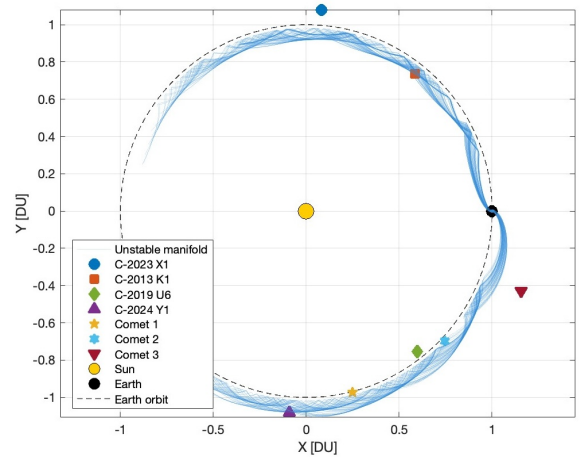


Fig. 13. Top view of the L_2 halo unstable manifolds.

Comet 1 and Comet 3, by contrast, cross the ecliptic outside the band swept by the manifolds and, despite lying in the same nominal quadrant, both benefit clearly from the swingby, which shortens the transfer by at least 150 days at any fixed ΔV .

These results provide direct quantitative support for the Comet Interceptor departure-strategy decision-making process: given the encounter geometry of a newly discovered target, the question of whether to commit to a lunar-flyby departure or to a direct escape can be addressed by locating the target within the encounter-geometry plane, and more precisely its ecliptic-crossing point relative to the manifold corridor.

IX. Acknowledgements

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